

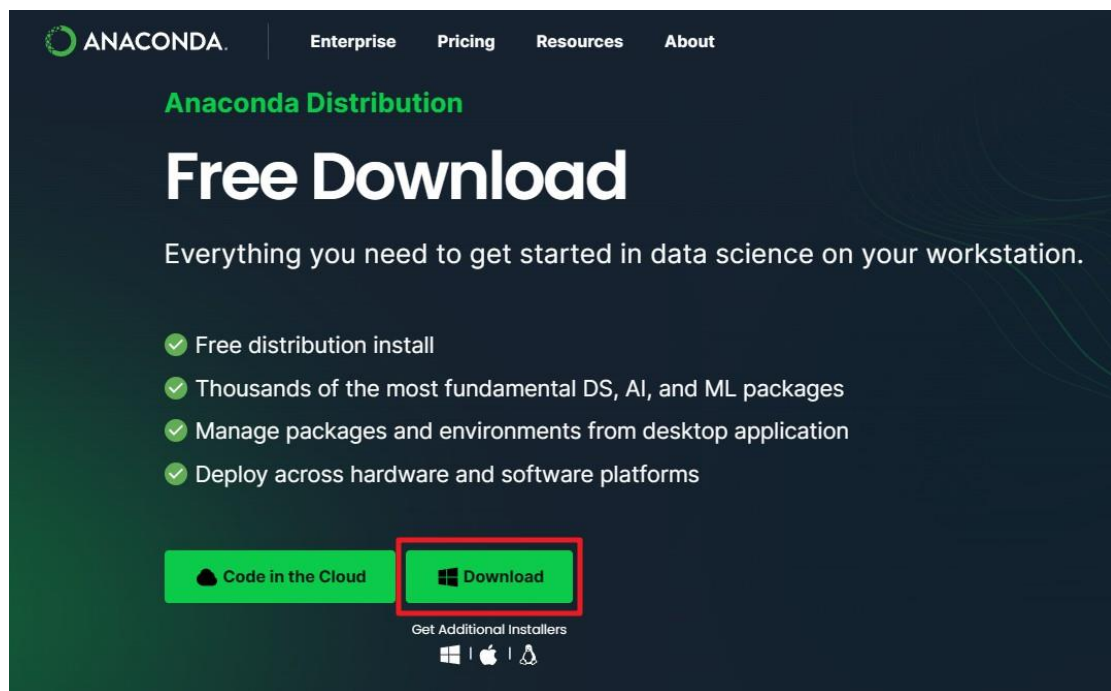
Anaconda + PyCharm + PyTorch (CPU) + 虚拟环境

原文链接: https://blog.csdn.net/weixin_44739914/article/details/129959139

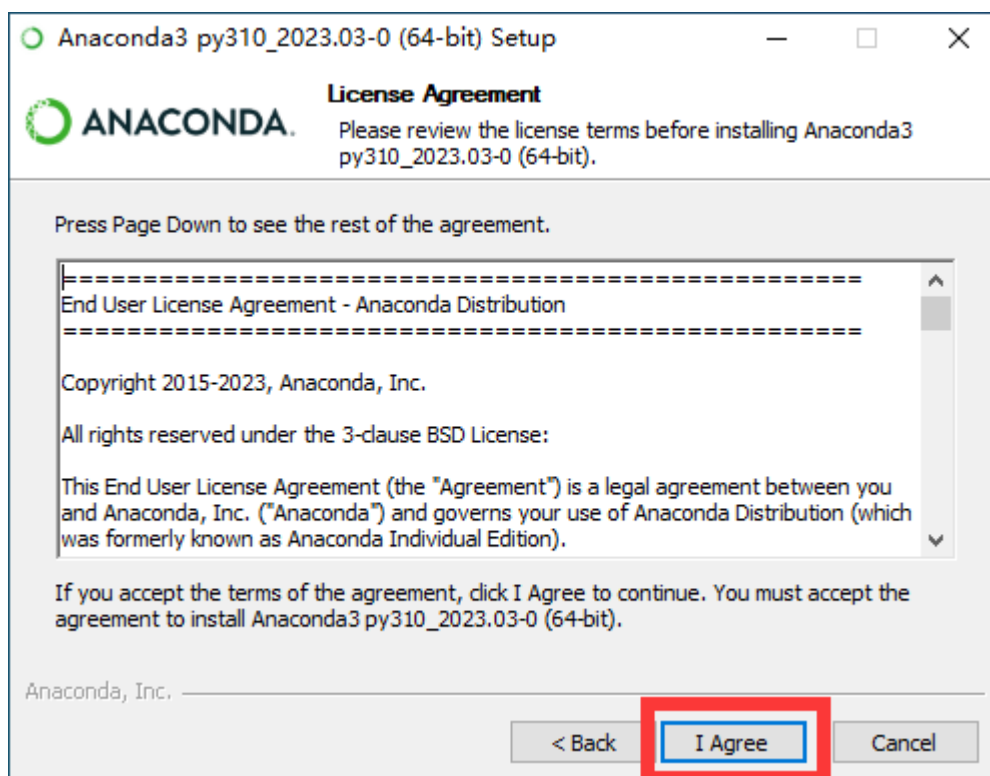
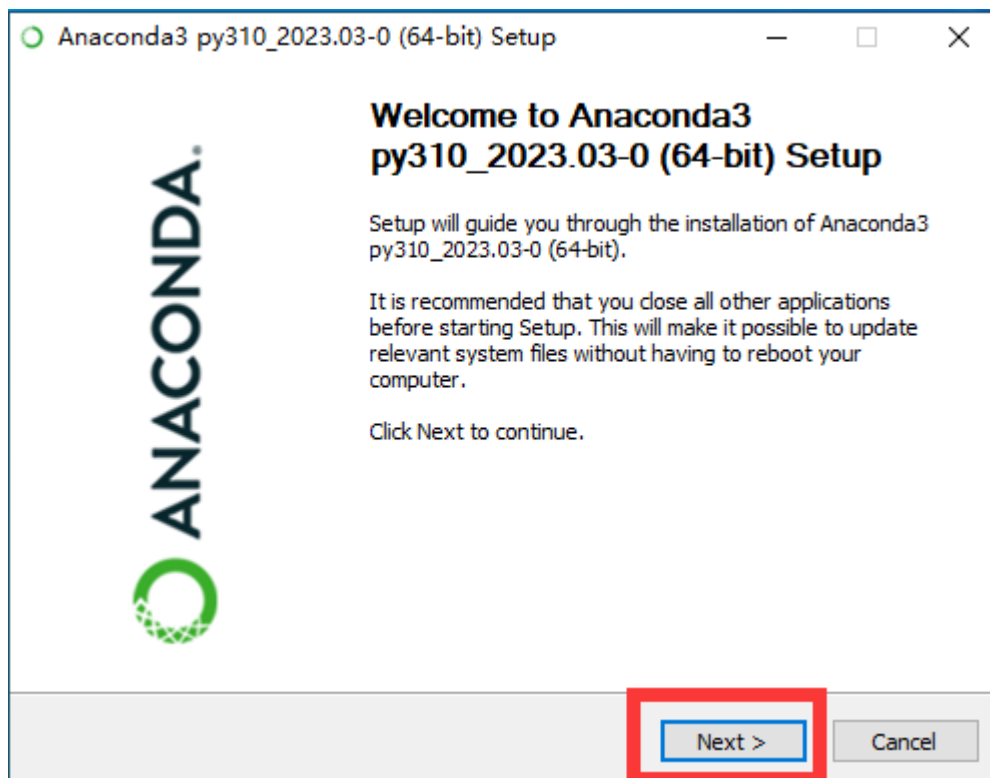
- 一、安装 Anaconda
- 二、安装 PyCharm
- 三、创建 虚拟环境 并 安装 PyTorch
- 四、关联 虚拟环境

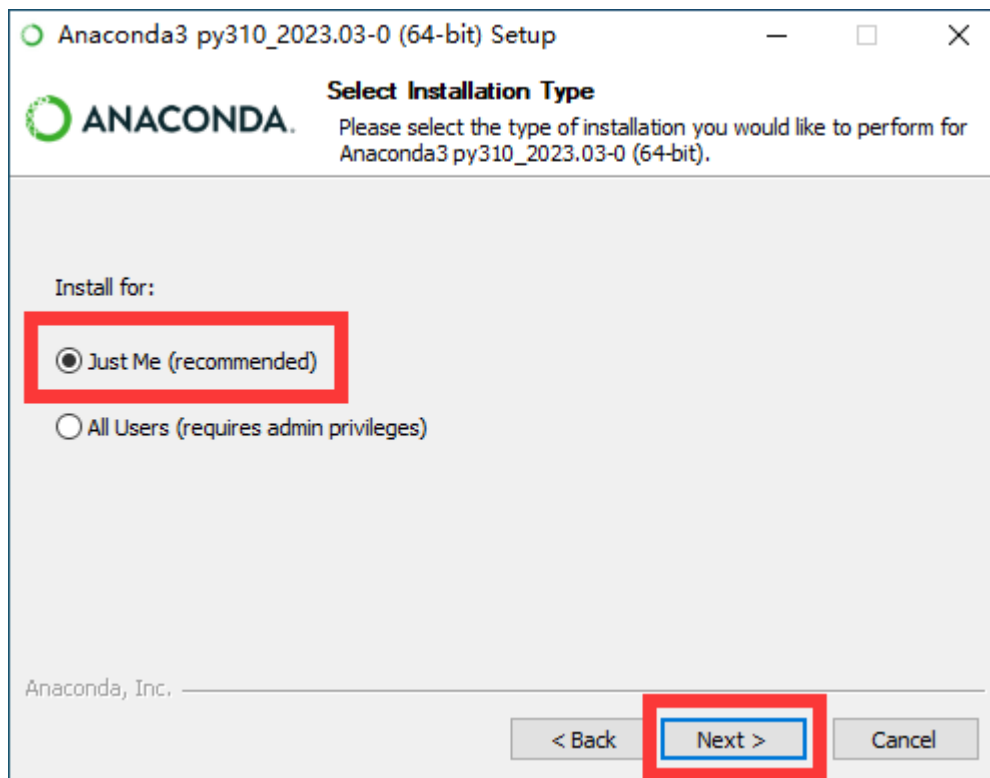
一、安装 Anaconda

进入 Anaconda 下载界面，单击 Download 下载。

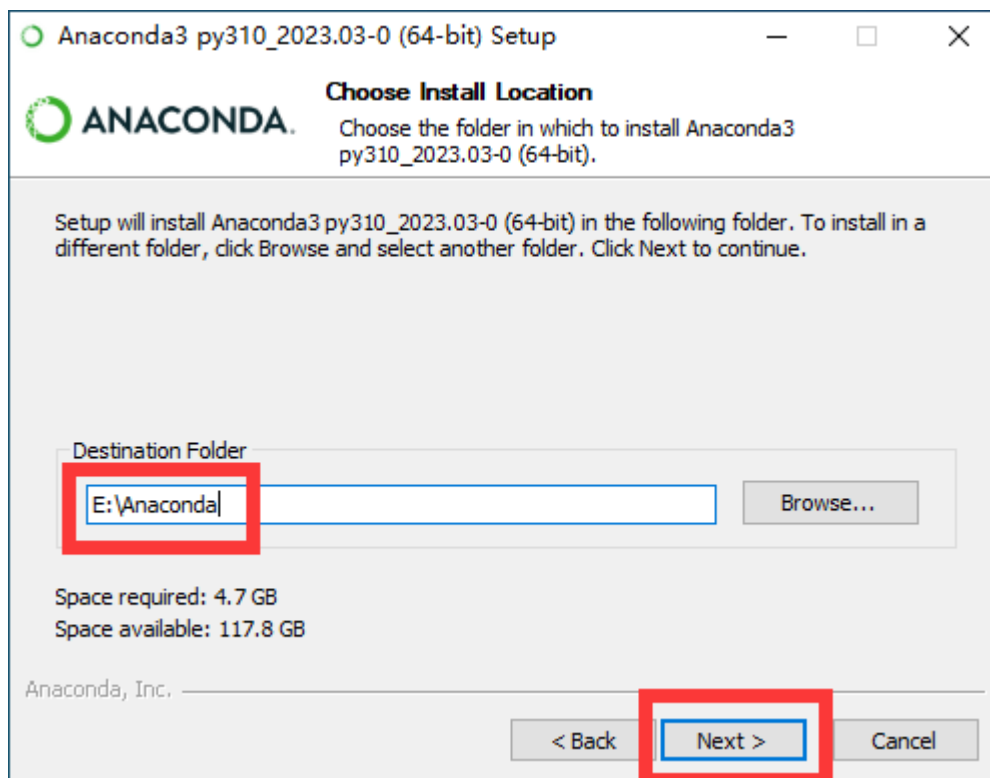


下载好之后，进行 安装。

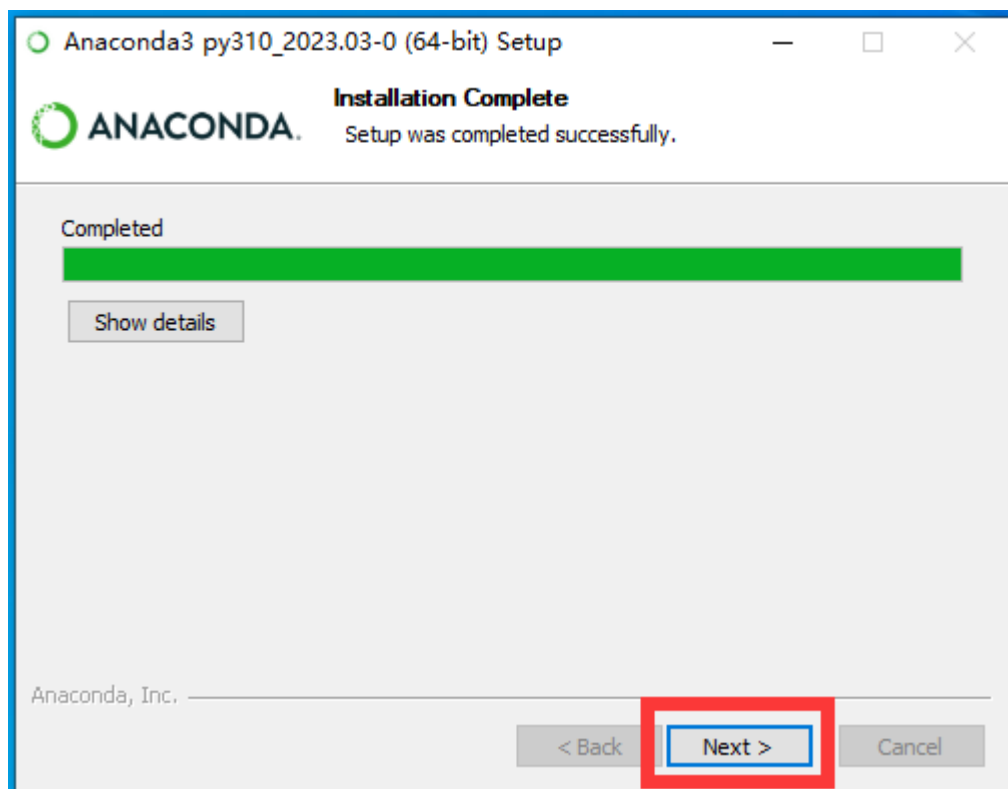
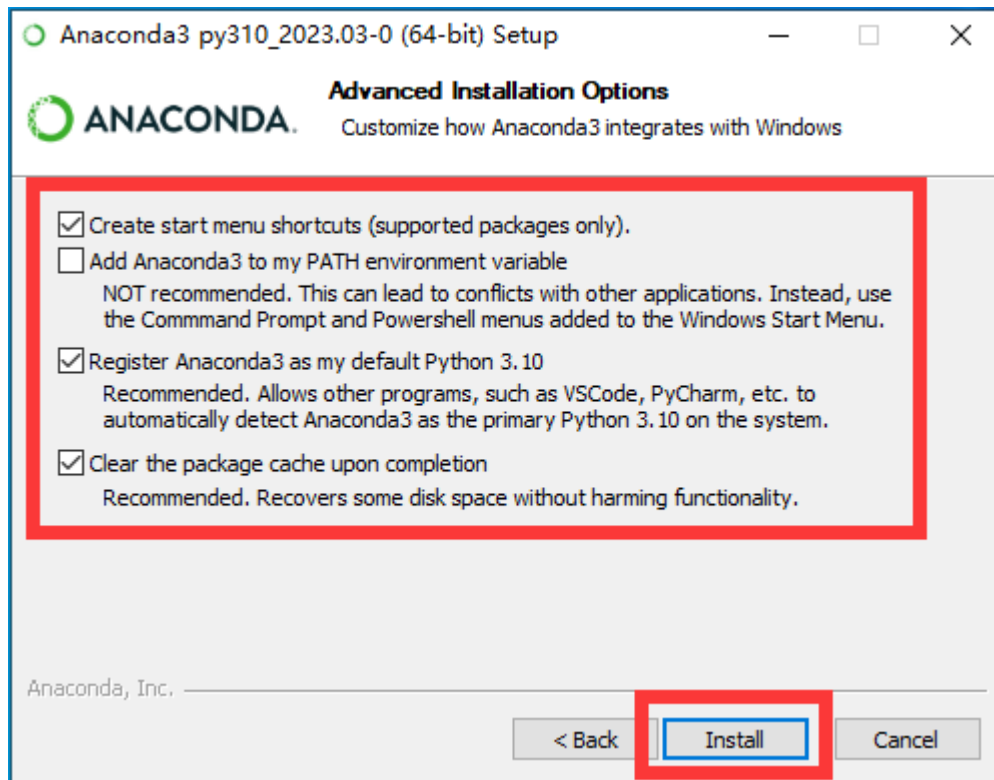


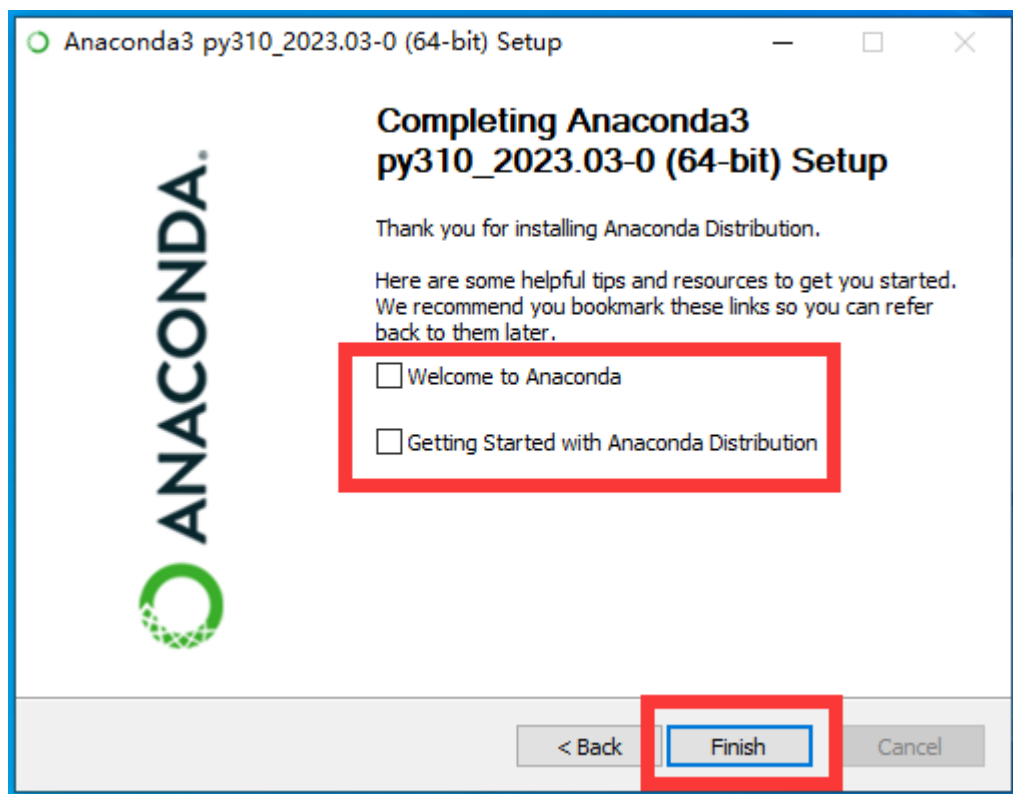
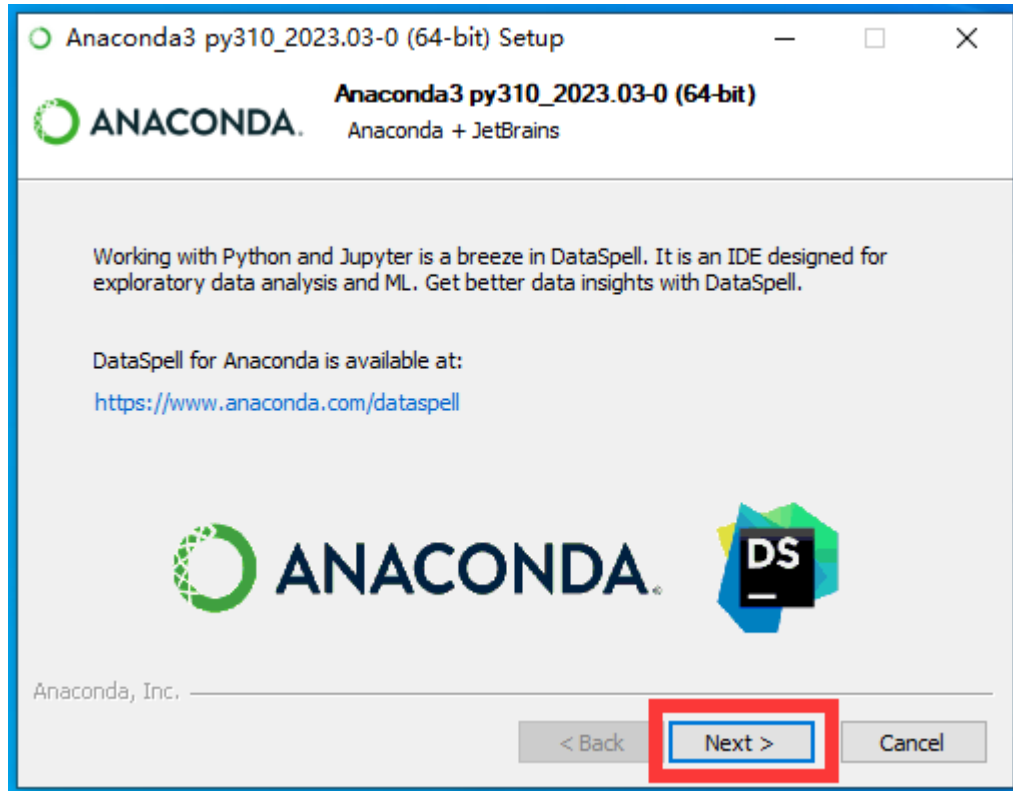


安装路径 全英文。

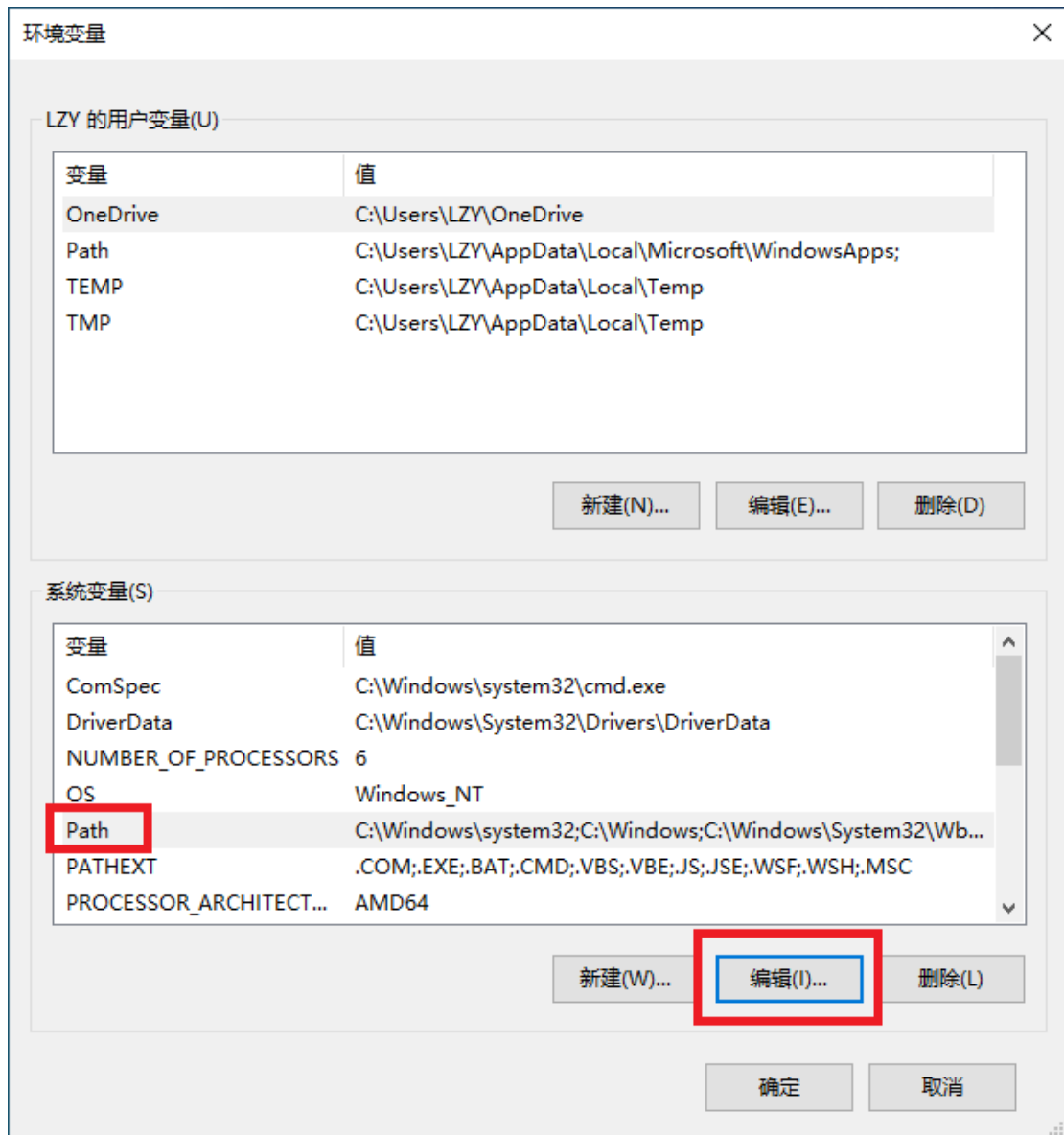


不勾选 第二项。

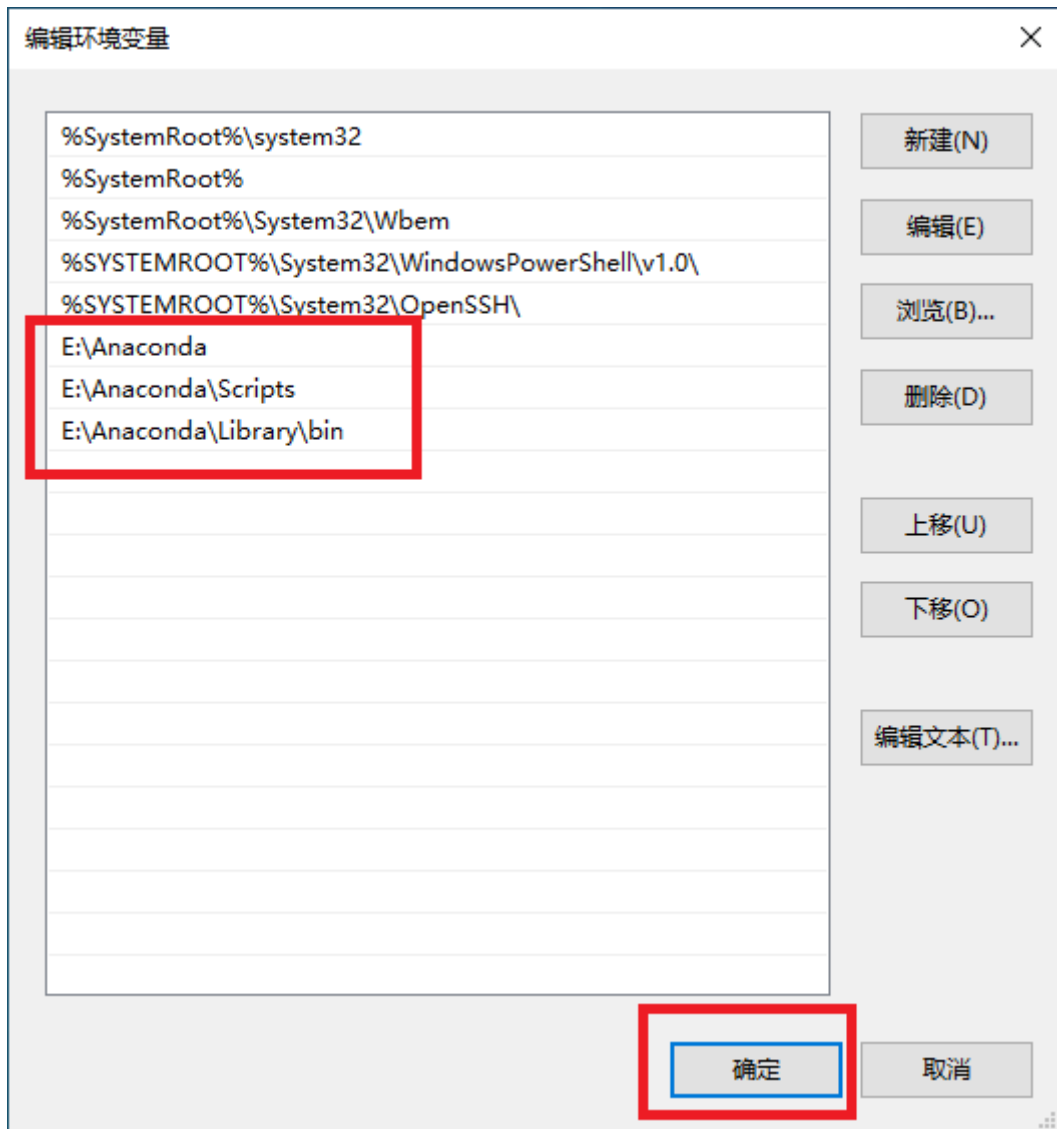




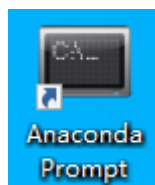
接下来配置 环境变量 。



基于自定义的 安装路径 ， 添加以下三个路径。



将以下图标从 开始菜单 中拖拽到 桌面 上，方便后续使用。



二、安装 PyCharm

首先，进入 [PyCharm 官网](https://www.jetbrains.com/pycharm/)，<https://www.jetbrains.com/pycharm/>，单击 **Download** 按钮 下载 [安装包](#)。



安装 **Community** 社区版（因为它免费），点击 **Download**，稍微等待即可下载。



Version: 2023.1
Build: 231.8109.197
30 March 2023

[System requirements](#)
[Installation instructions](#)
[Other versions](#)
[Third-party software](#)

Download PyCharm

[Windows](#) [macOS](#) [Linux](#)

Professional

For both Scientific and Web Python development. With HTML, JS, and SQL support.

[Download](#) [.exe](#) ▼

Free 30-day trial available

Community

For pure Python development

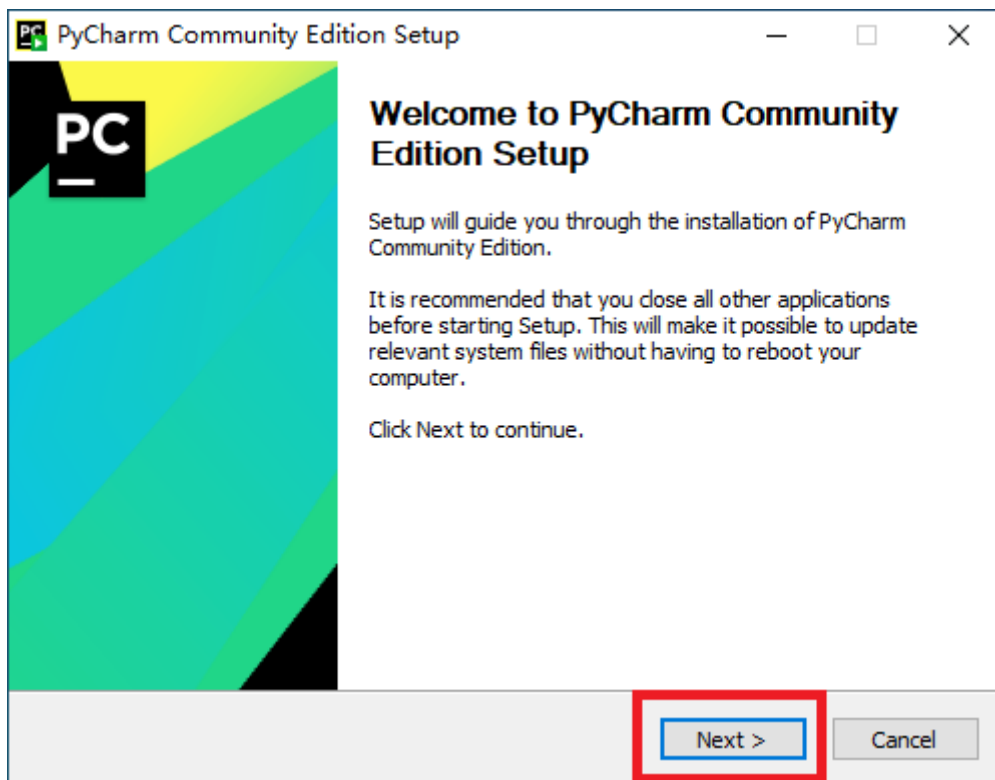
[Download](#) [.exe](#) ▼

Free, built on open-source

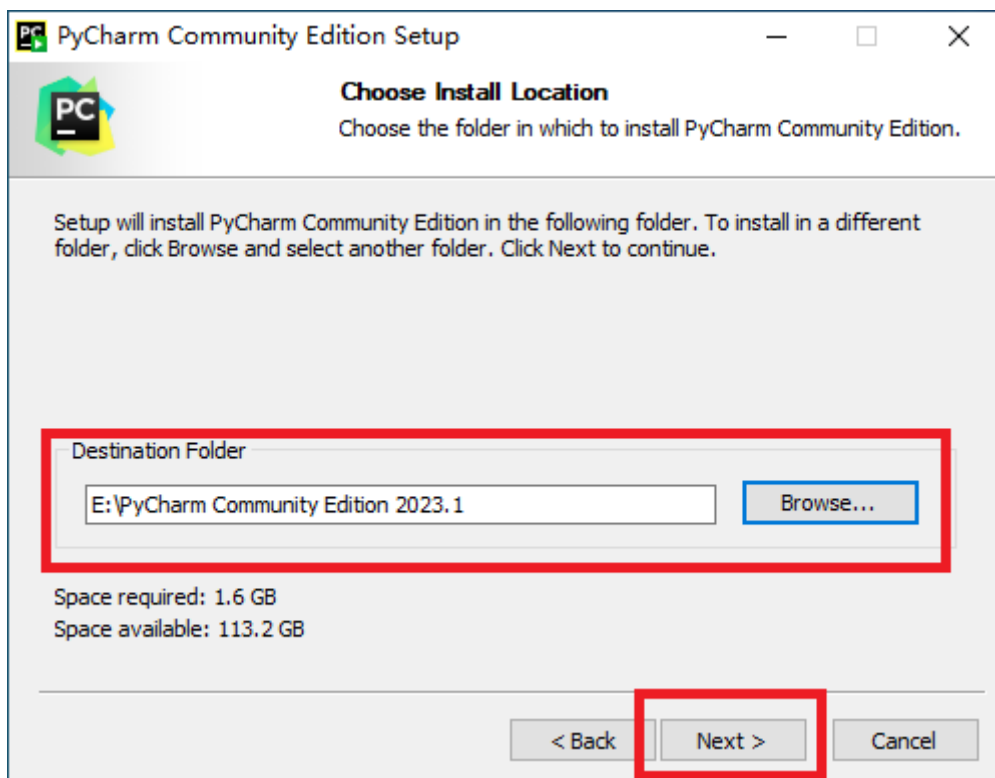


Get the Toolbox App to download PyCharm and its future updates with ease

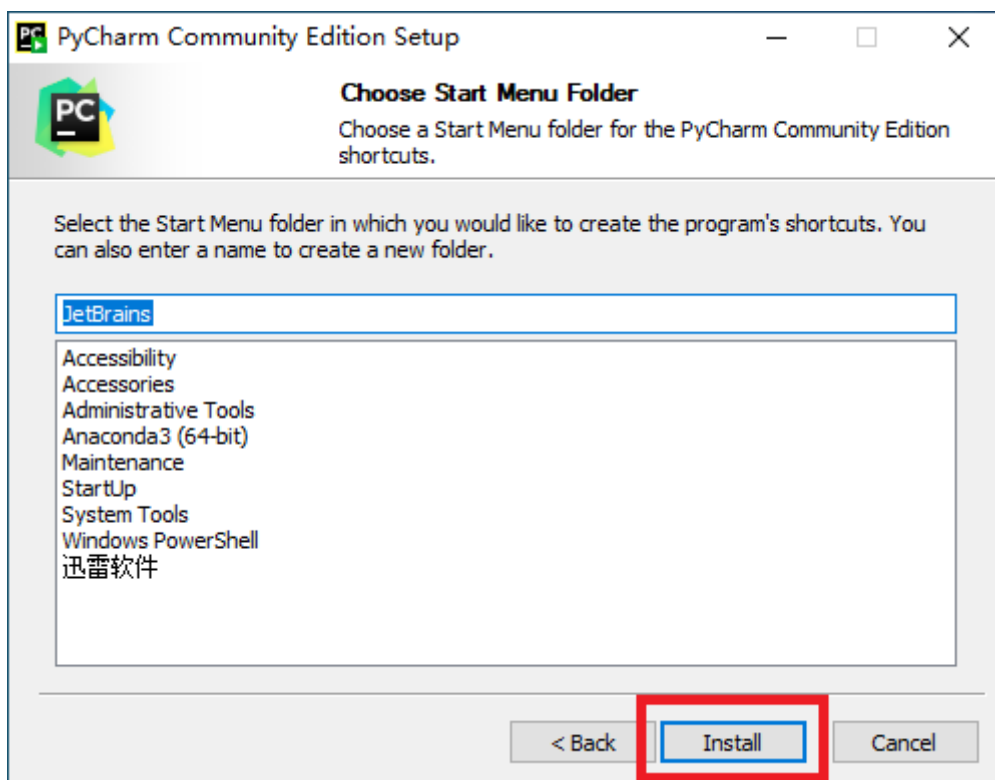
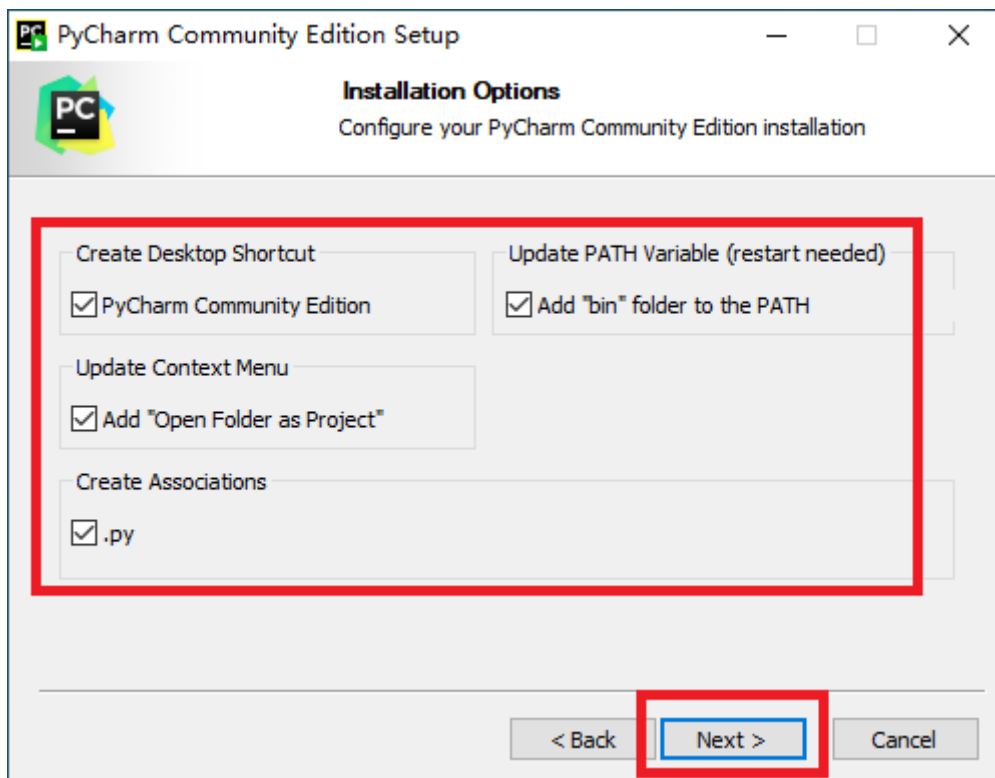
下载完成后，双击打开安装包，进行 **安装** 操作。



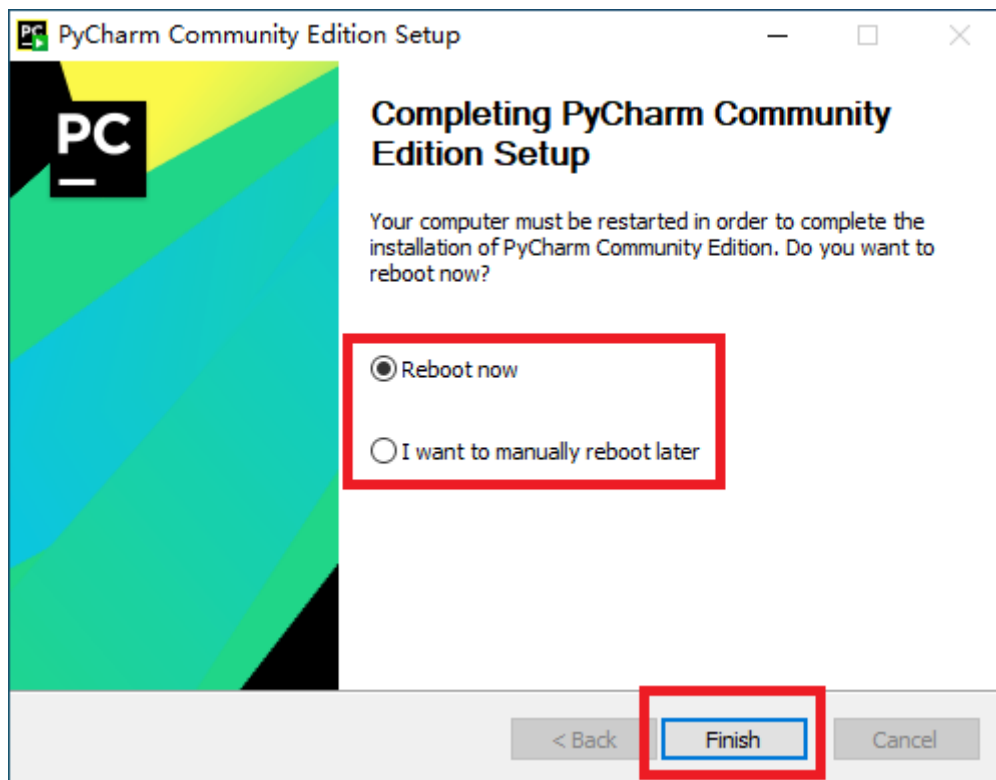
自定义一个 安装路径，最好能做到“见名知意”。



无脑 全选，都用得到。

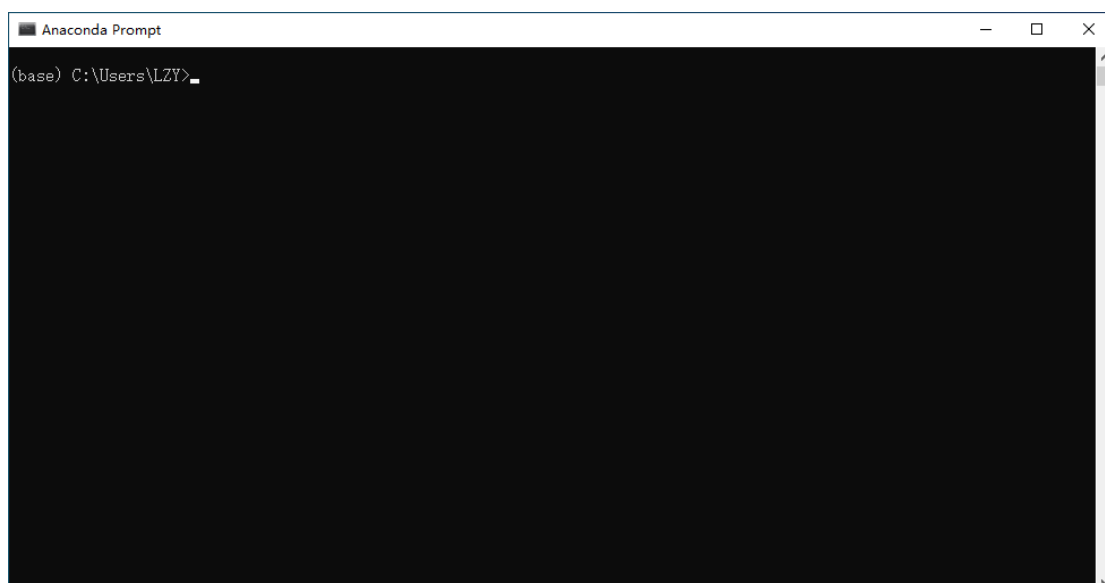
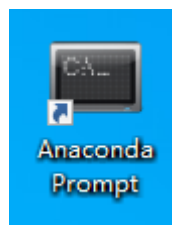


想 重启 电脑，就勾选 第一项，不想重启，就勾选第二项。



三、创建 虚拟环境 并 安装 PyTorch

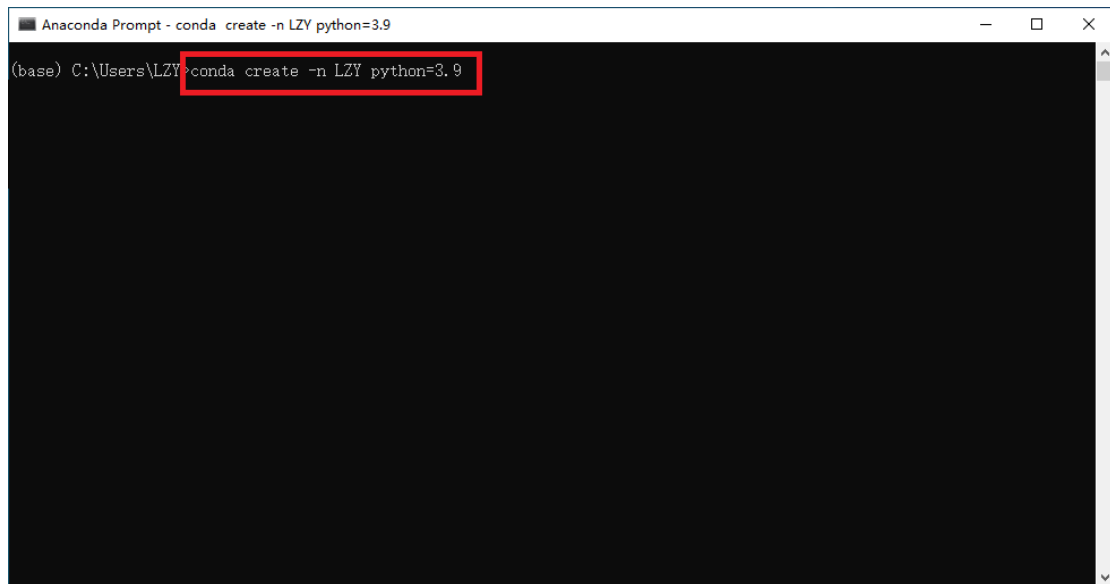
打开 **Anaconda Prompt** 。



创建 虚拟环境（虚拟环境名自己设置）。

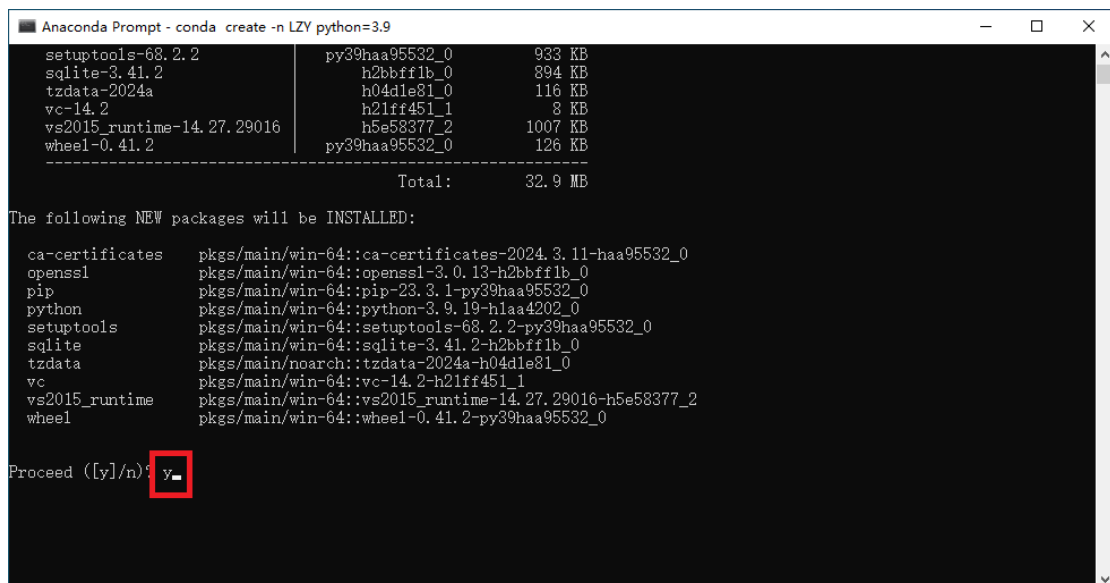
conda create -n 虚拟环境名 python=python 版本

conda create -n BNUAI python=3.11



```
Anaconda Prompt - conda create -n LZY python=3.9
(base) C:\Users\LZY>conda create -n LZY python=3.9
```

在出现一大段英文之后，输入字母 **y**，再按下回车。



```
Anaconda Prompt - conda create -n LZY python=3.9
setuptools-68.2.2      py39haa95532_0      933 KB
sqlite-3.41.2         h2bbff1b_0          894 KB
tzdata-2024a         h04d1e81_0          116 KB
vc-14.2               h21ff451_1           8 KB
vs2015_runtime-14.27.29016 h5e58377_2        1007 KB
wheel-0.41.2          py39haa95532_0       126 KB
-----
Total:                 32.9 MB

The following NEW packages will be INSTALLED:

ca-certificates      pkgs/main/win-64::ca-certificates-2024.3.11-haa95532_0
openssl              pkgs/main/win-64::openssl-3.0.13-h2bbff1b_0
pip                  pkgs/main/win-64::pip-23.3.1-py39haa95532_0
python               pkgs/main/win-64::python-3.9.19-h1aa4202_0
setuptools            pkgs/main/win-64::setuptools-68.2.2-py39haa95532_0
sqlite               pkgs/main/win-64::sqlite-3.41.2-h2bbff1b_0
tzdata               pkgs/main/noarch::tzdata-2024a-h04d1e81_0
vc                   pkgs/main/win-64::vc-14.2-h21ff451_1
vs2015_runtime       pkgs/main/win-64::vs2015_runtime-14.27.29016-h5e58377_2
wheel                pkgs/main/win-64::wheel-0.41.2-py39haa95532_0

Proceed ([y]/n)? y
```

```
Anaconda Prompt

Proceed ([y]/n)? y

Downloading and Extracting Packages:
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#   $ conda activate LZY
#
# To deactivate an active environment, use
#
#   $ conda deactivate

(base) C:\Users\LZY>
```

激活 虚拟环境。

```
conda activate 虚拟环境名
```

```
conda activate BNUAI
```

创建 PyTorch 环境之前，先安装一些常用的包，可以提高后续的速度。

```
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple numpy
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple pandas
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple pillow==9.5.0
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple matplotlib
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple tqdm
```

创建 **PyTorch** 环境，如果失败就多试几次，有可能是 **网络原因**。

2024 年 4 月 6 日 13 点 28 分 再次测试 依然能用！

2024 年 5 月 11 日 16 点 58 分 再次测试 依然能用！

(1) **cpu** 版本的安装命令，其他版本可参考官网：

```
pip install torch torchvision torchaudio
```

```
(pytorch) C:\Users\Administrator>pip install torch torchvision torchaudio
Collecting torch
  Downloading torch-1.13.0-cp310-cp310-win_amd64.whl (167.3 MB)
    167.3/167.3 MB 1.4 MB/s eta 0:00:00
Collecting torchvision
  Downloading torchvision-0.14.0-cp310-cp310-win_amd64.whl (1.1 MB)
    1.1/1.1 MB 1.1 MB/s eta 0:00:00
Collecting torchaudio
  Downloading torchaudio-0.13.0-cp310-cp310-win_amd64.whl (2.1 MB)
    2.1/2.1 MB 1.2 MB/s eta 0:00:00
Collecting typing-extensions
  Downloading typing_extensions-4.4.0-py3-none-any.whl (26 kB)
Collecting requests
  Downloading requests-2.28.1-py3-none-any.whl (62 kB)
    62.8/62.8 kB 3.3 MB/s eta 0:00:00
Collecting pillow<8.3.*,>=5.3.0
  Downloading Pillow-9.3.0-cp310-cp310-win_amd64.whl (2.5 MB)
    2.5/2.5 MB 1.5 MB/s eta 0:00:00
Collecting numpy
  Downloading numpy-1.23.4-cp310-cp310-win_amd64.whl (14.6 MB)
    14.6/14.6 MB 1.6 MB/s eta 0:00:00
Collecting charset-normalizer<3,>=2
  Downloading charset_normalizer-2.1.1-py3-none-any.whl (39 kB)
Collecting urllib3<1.27,>=1.21.1
  Downloading urllib3-1.26.12-py2.py3-none-any.whl (140 kB)
    140.4/140.4 kB 1.4 MB/s eta 0:00:00
Requirement already satisfied: certifi>=2017.4.17 in c:\users\administrator\conda\envs\pytorch\lib\site-packages (from requests->torchvision) (2022.9.24)
Collecting idna<4,>=2.5
  Downloading idna-3.4-py3-none-any.whl (61 kB)
    61.5/61.5 kB 1.1 MB/s eta 0:00:00
Installing collected packages: urllib3, typing-extensions, pillow, numpy, idna, charset-normalizer, torch, requests, torchvision, torchaudio
Successfully installed charset-normalizer-2.1.1 idna-3.4 numpy-1.23.4 pillow-9.3.0 requests-2.28.1 torch-1.13.0 torchaudio-0.13.0 torchvision-0.14.0 typing-extensions-4.4.0
```

验证 Pytorch 是否安装成功;

```
>>> print(torch.__version__)
1.13.0+cpu
```

查看 pytorch 版本: 若出现则说明安装成功。

```
import torch
print(torch.__version__)
```

注意: 是两个下划线。

再次验证是否成功安装:

```
>>> import torch
>>> x=torch.rand(5,3)
>>> print(x)
tensor([[0.5618, 0.2708, 0.5189],
        [0.0109, 0.4416, 0.2155],
        [0.7985, 0.0351, 0.4655],
        [0.4963, 0.9962, 0.6830],
        [0.7332, 0.8236, 0.4314]])
```

```
import torch
x = torch.rand(5,3)
```

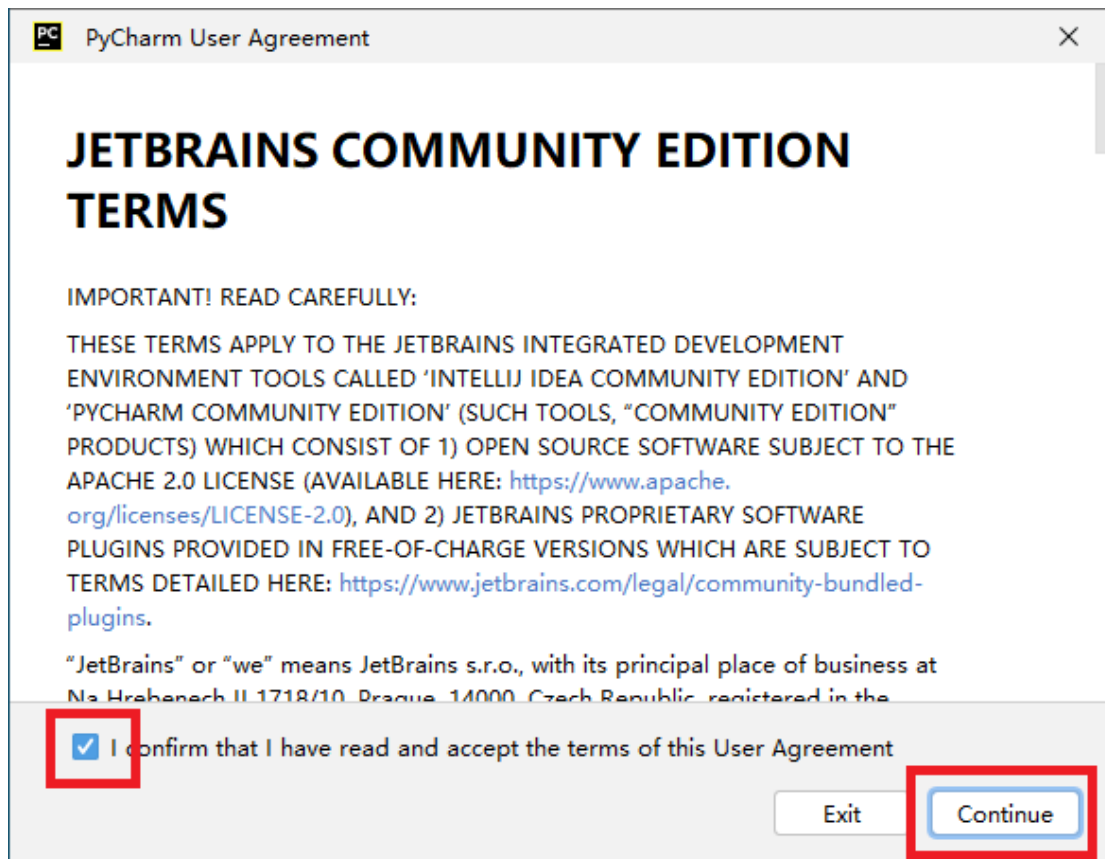
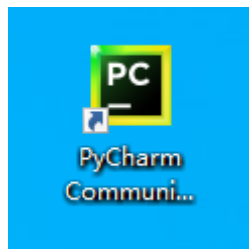
(2) GPU 支持的 Pytorch 安装

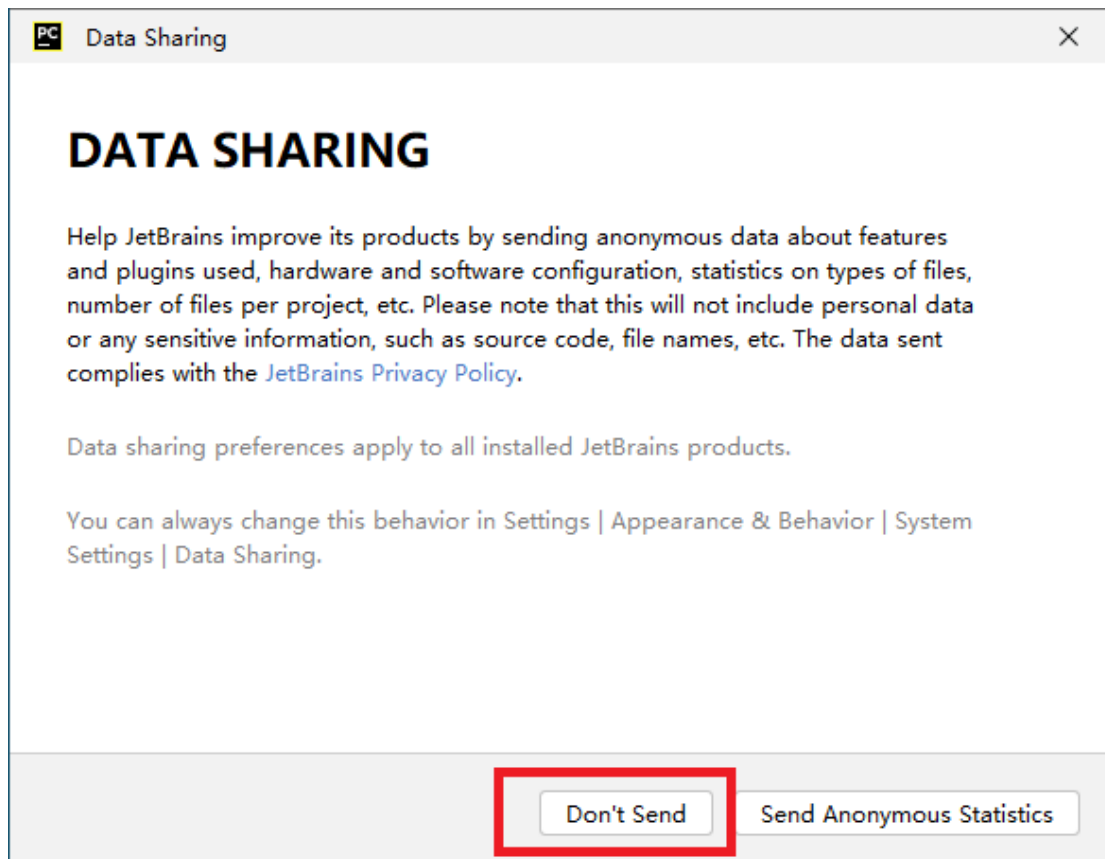
```
pip install torch==1.12.0+cu113 torchvision==0.13.0+cu113 torch
audio==0.12.0 -f https://download.pytorch.org/whl/torch_stable.htm
l
```

```
Anaconda Prompt
(LZY) C:\Users\LZY>pip install torch==1.12.0+cu113 torchvision==0.13.0+cu113 torchaudio==0.12.0 -f https://download.pytorch.org/whl/torch_stable.html
Looking in links: https://download.pytorch.org/whl/torch_stable.html
Requirement already satisfied: torch==1.12.0+cu113 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (1.12.0+cu113)
Requirement already satisfied: torchvision==0.13.0+cu113 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (0.13.0+cu113)
Requirement already satisfied: torchaudio==0.12.0 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (0.12.0+cu113)
Requirement already satisfied: typing-extensions in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from torch==1.12.0+cu113) (4.11.0)
Requirement already satisfied: numpy in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from torchvision==0.13.0+cu113) (1.26.4)
Requirement already satisfied: requests in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from torchvision==0.13.0+cu113) (2.31.0)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from torchvision==0.13.0+cu113) (9.5.0)
Requirement already satisfied: charset-normalizer<4,>=2 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from requests->torchvision==0.13.0+cu113) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from requests->torchvision==0.13.0+cu113) (3.6)
Requirement already satisfied: urllib3<3,>=1.21.1 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from requests->torchvision==0.13.0+cu113) (2.2.1)
Requirement already satisfied: certifi>=2017.4.17 in d:\software\anaconda\anaconda\envs\lzy\lib\site-packages (from requests->torchvision==0.13.0+cu113) (2024.2.2)
(LZY) C:\Users\LZY>
```

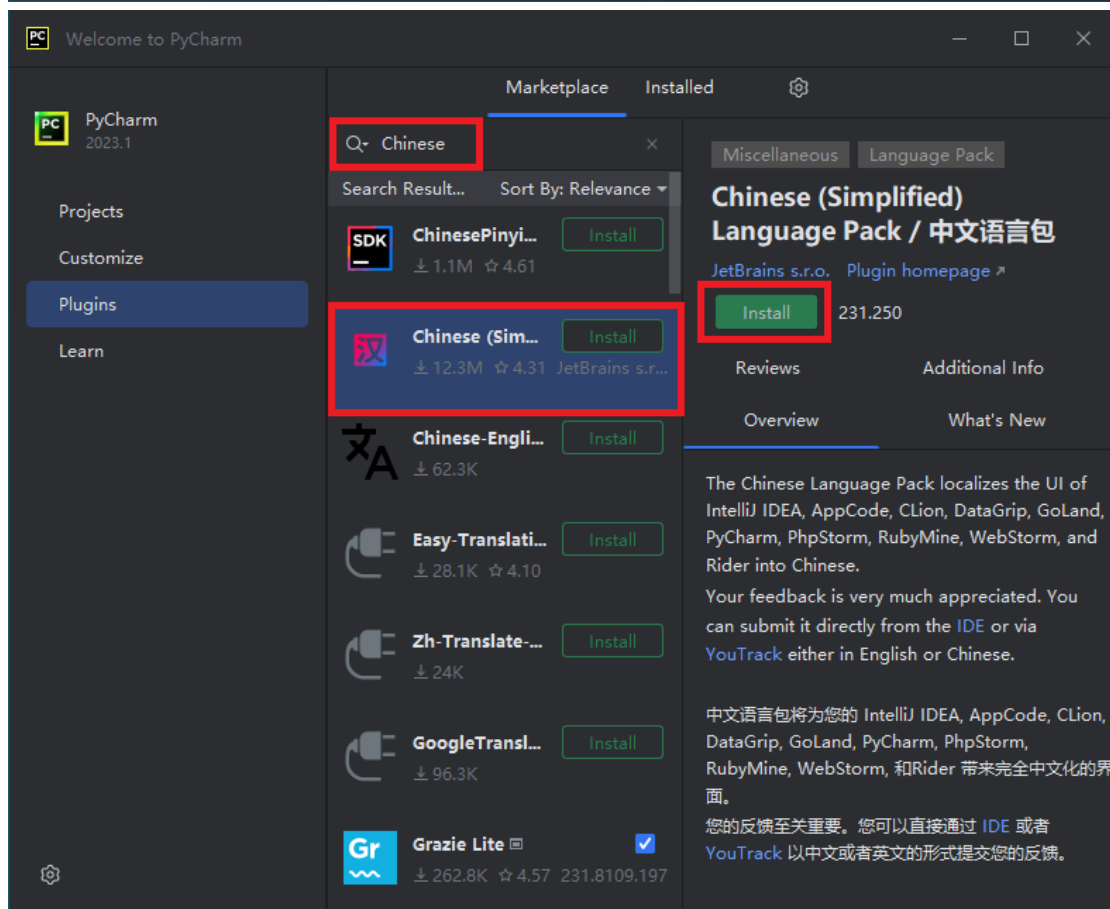
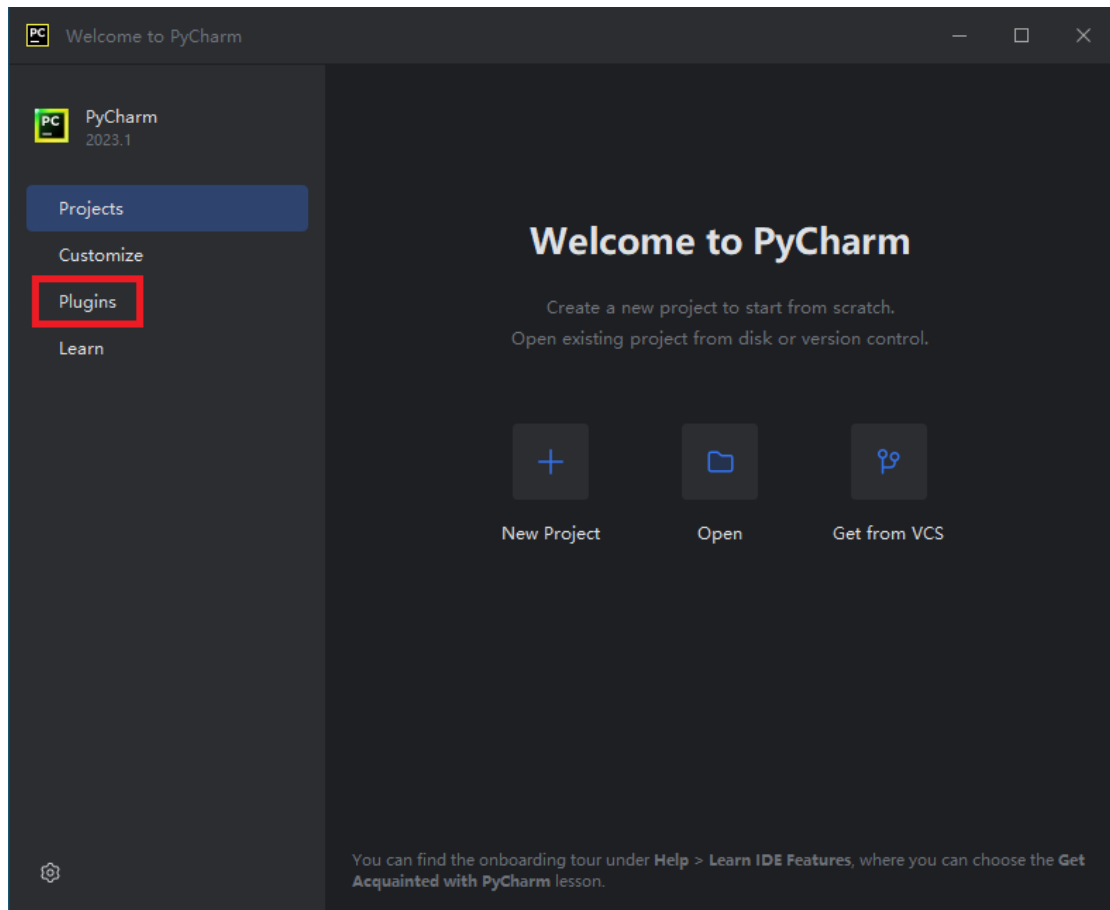
四、关联 虚拟环境

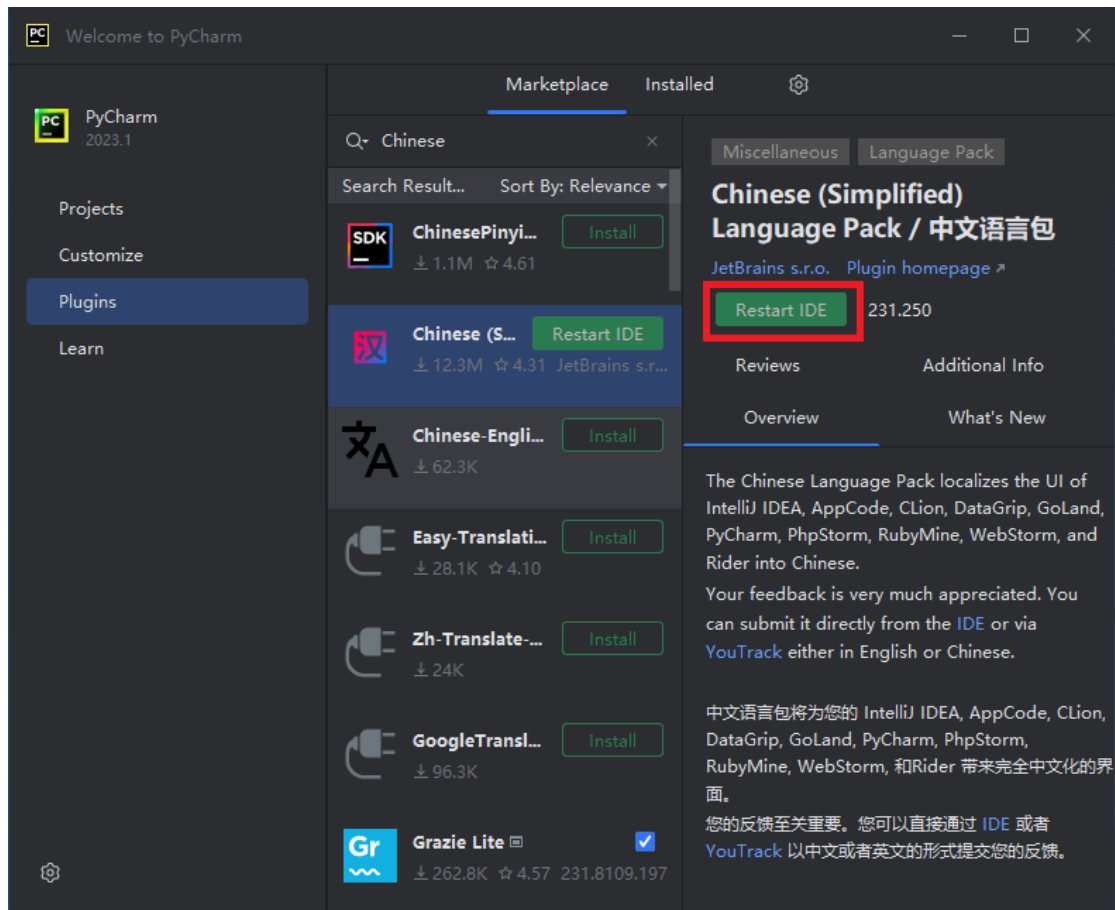
双击打开桌面上的 PyCharm 。



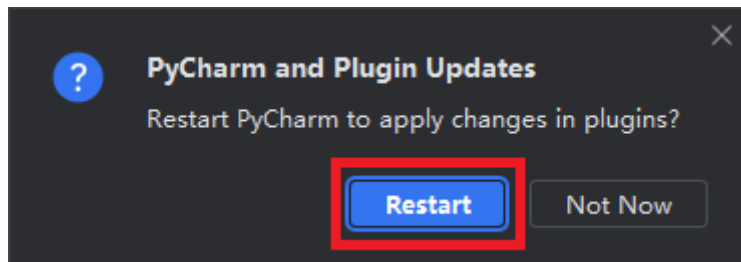


如你所见，英文的，要换成 中文 。

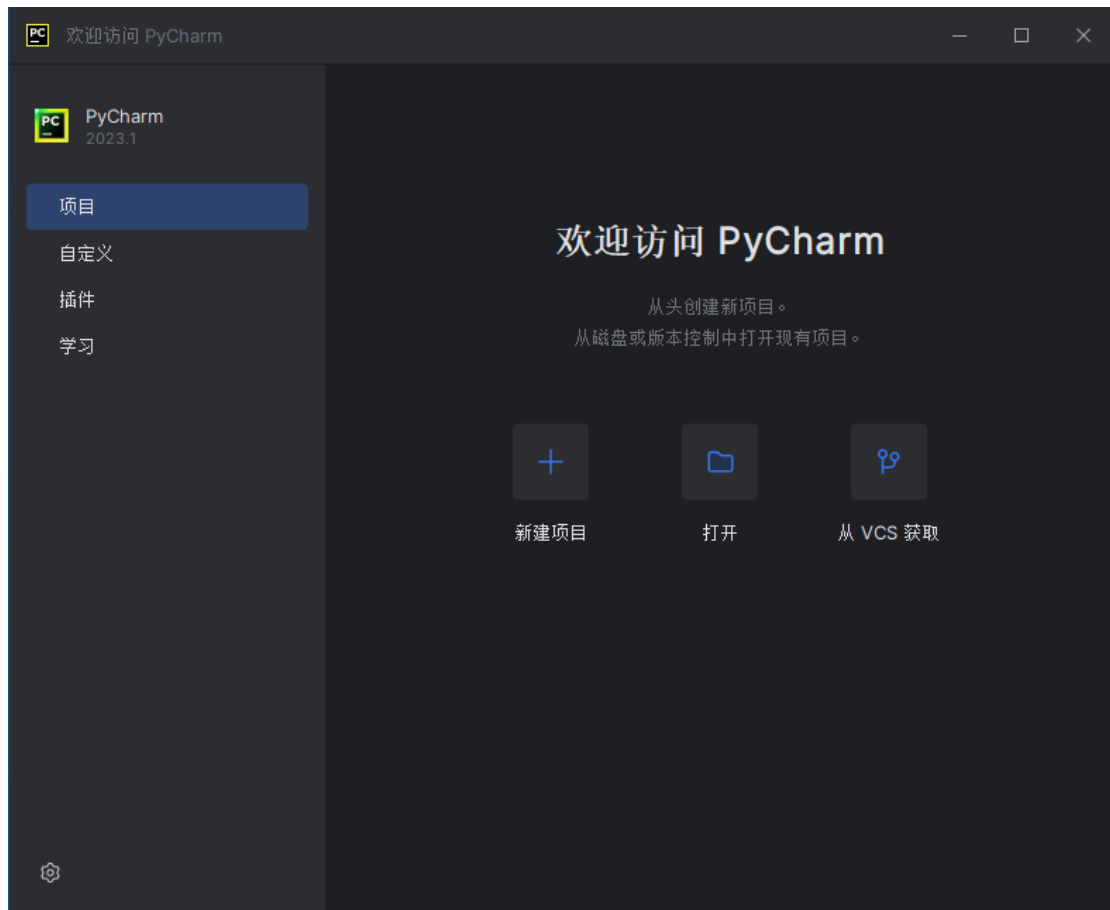




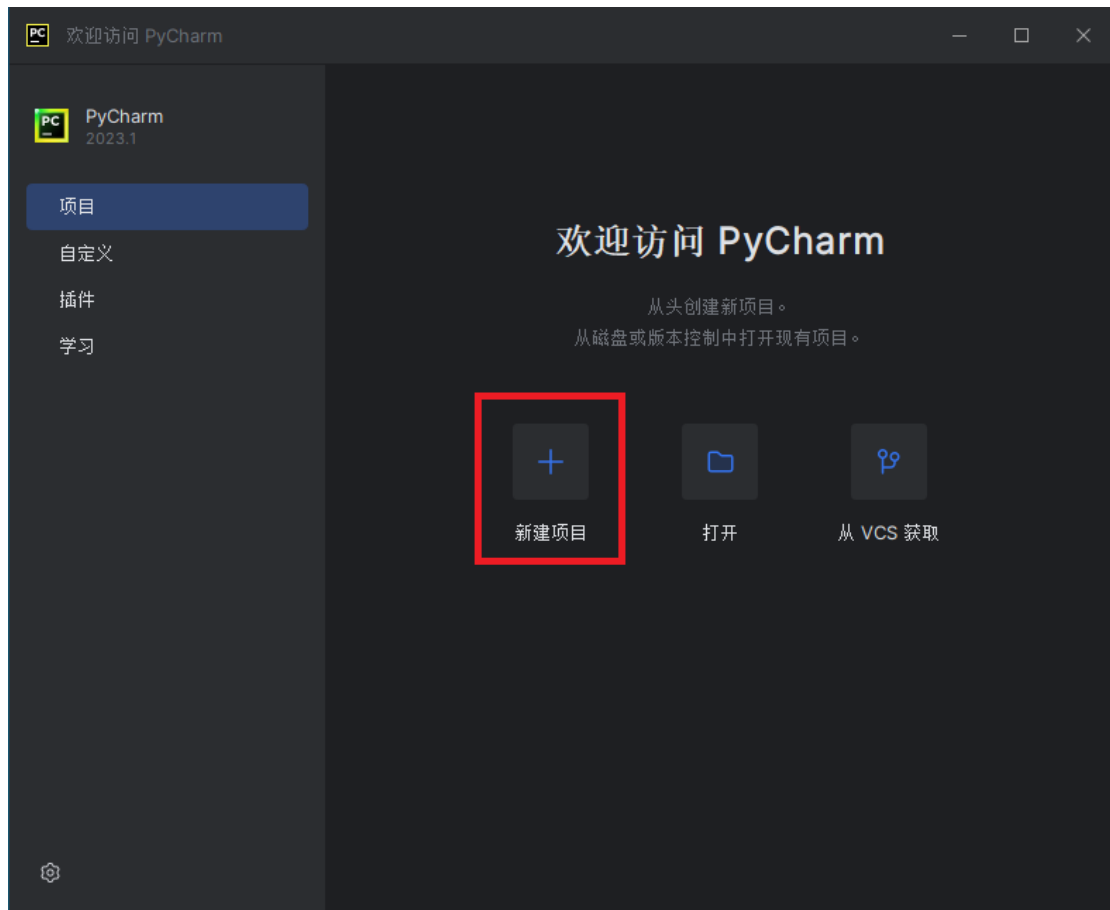
点击 Restart IDE 重启一下 PyCharm 。



终于，重启之后，熟悉的 中文 界面出来了。



接下来就是让 PyCharm 与 Anaconda 中的 虚拟环境 关联起来，点击 新建项目 。



项目的位置可以自己设置。

应该会有两个 环境 。

第一个环境是 `base`（基准） 环境，别动它！

选择第二个 自己创建的虚拟环境 ， 点击 创建 。



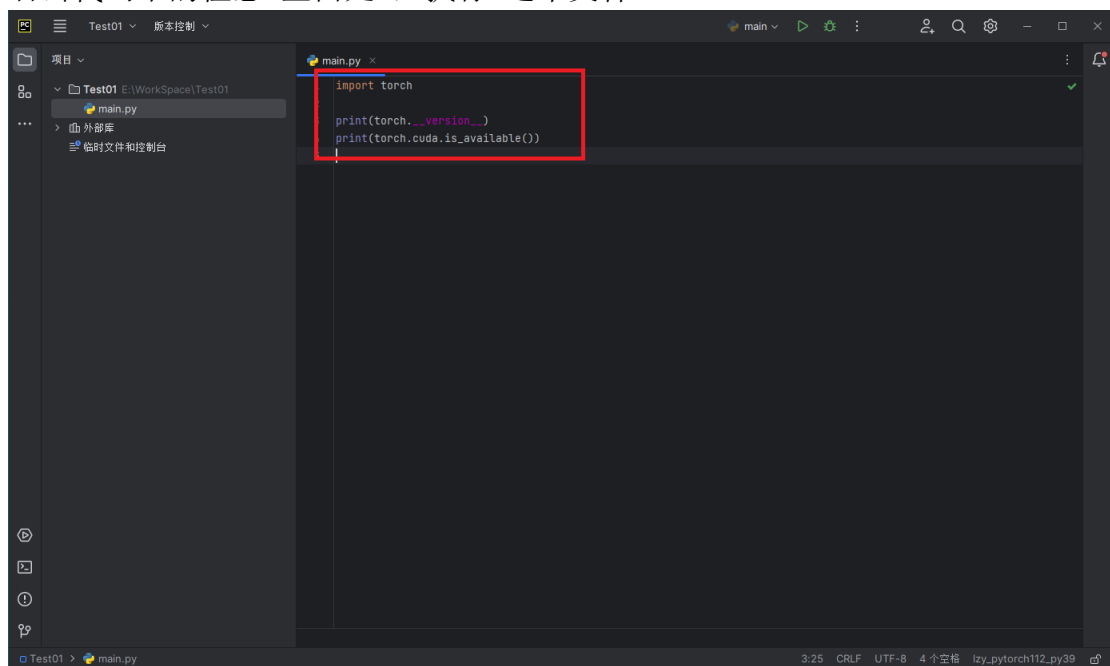
创建 .py 文件，输入以下代码进行测试。

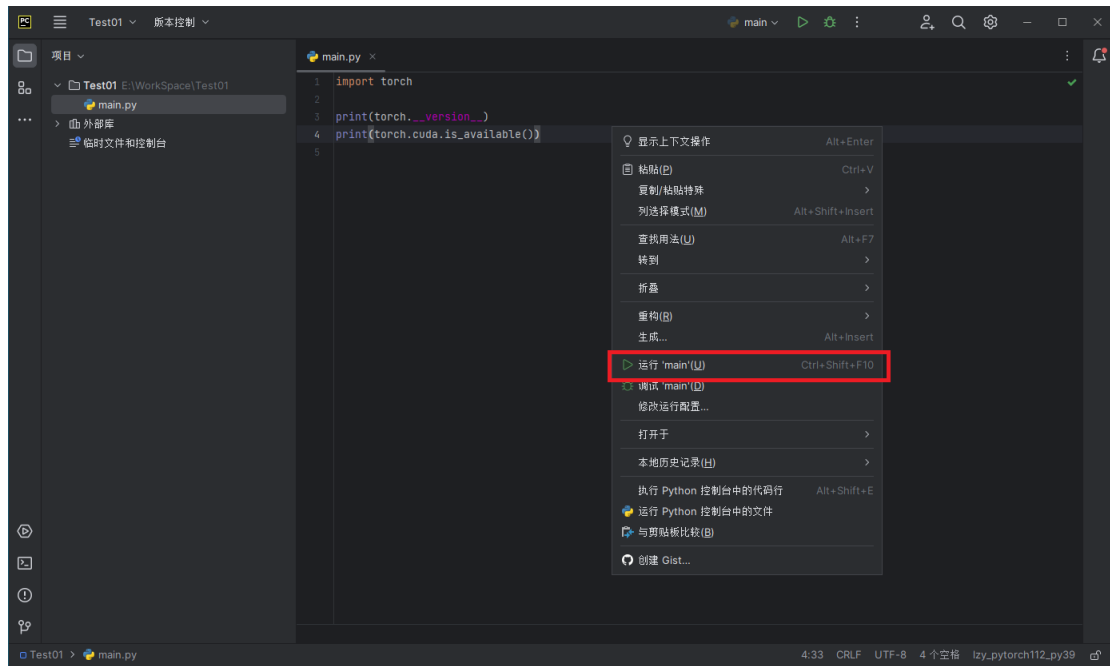
```
import torch
```

```
print(torch.__version__)
```

```
print(torch.cuda.is_available())
```

右击代码中的任意 空白处，执行 这个文件。





最终结果如下，说明已经成功安装好 PyTorch ，可以开心的继续学习了。

```
1.12.0+cu113
True

进程已结束，退出代码为 0
```